NOVA HORIZON

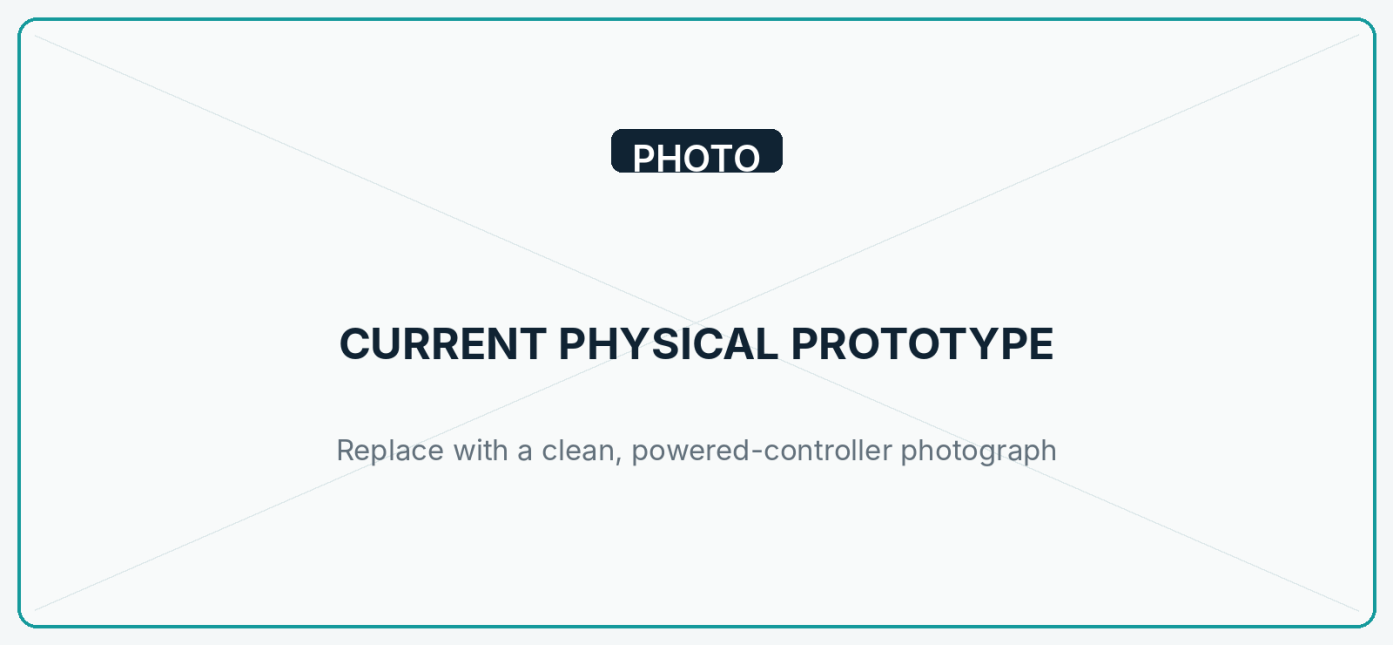
CONTROLLER

A local-first touchscreen controller prototype for monitoring and cautiously controlling Bambu Lab printers.

INDEPENDENT BAMBU LAB CONTROLLER DEVELOPMENT PROJECT

Prepared by Dima

Firmware reviewed: v0.7.1 deployed prototype build / July 2026



Current physical prototype.

Bambu Lab is not currently affiliated with or responsible for this independent project.

1 PROJECT OVERVIEW

Nova Horizon Controller is a dedicated touchscreen device being developed to put useful printer information and carefully gated controls on a small, always-available local display. The current prototype targets a Bambu Lab P1S and is intended to reduce the need to reach for a phone or computer for routine at-a-glance monitoring and selected printer actions.

The project is local-first: the working P1S integration communicates over the local network. The design leaves clear boundaries for optional Nova Horizon web services and a Jetson Orin Nano vision system, but neither is part of the current working printer path.

2 CURRENT DEVELOPMENT STATUS

STATUS	CURRENT EVIDENCE-BASED POSITION
Completed and verified	Firmware build; 4.3-inch display; landscape LVGL UI; aligned touch; ESP32-C6 Wi-Fi; local P1S authentication; live print reporting; chamber-light control; P1 camera liveview; physically enabled LAN OTA.
Functional, refining	P1S telemetry, on-screen setup, reconnect behavior, SD mounting, camera lifecycle, and UI messaging. Signed v0.7.1 is deployed with stable short health samples; full regression remains pending.
In active development	Safe validation of pause/resume/cancel, credential-storage hardening, local file indexing and print initiation, and broader regression tests.
Planned	AMS/filament workflows, multi-printer profiles, optional Nova Horizon backend connectivity, Jetson vision verification, and broader diagnostics.
Needs more printers	X- and H-series authentication, telemetry, cameras, AMS generations, command behavior, firmware differences, errors, and UI adaptation.



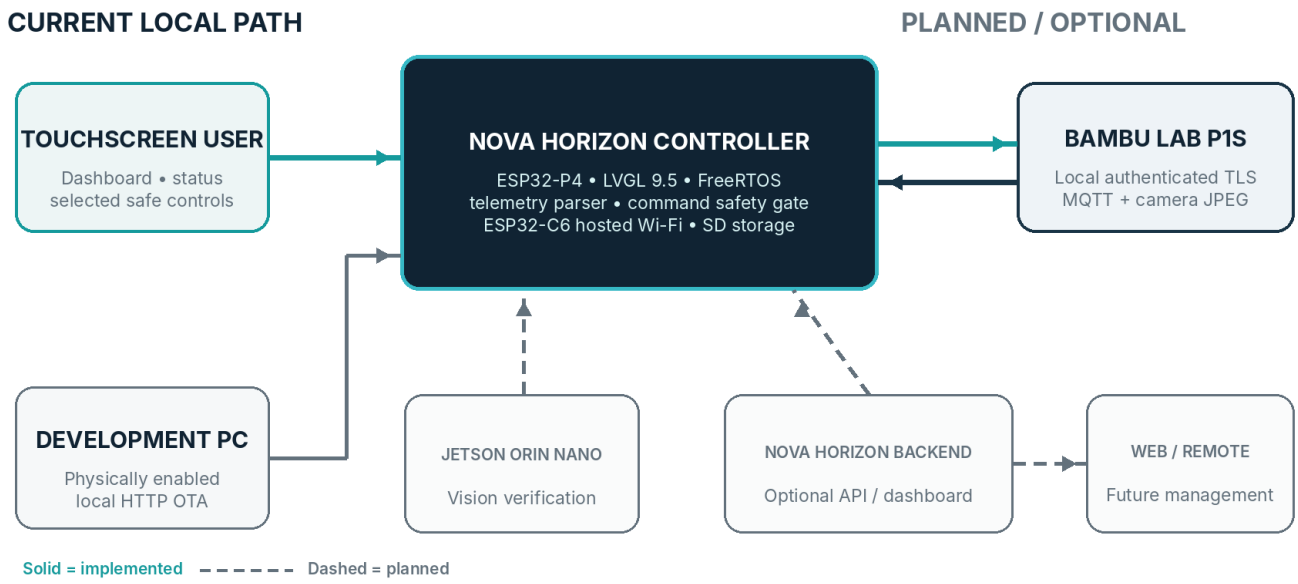
Current touchscreen interface.

3 KEY PLANNED CAPABILITIES

CAPABILITY	STATUS	CURRENT BOUNDARY
Status / monitoring	Working on P1S, refining	Connection and job state, progress, layers, remaining time, temperatures, fan level, and chamber-light state.
Job management	Implemented; validation pending	Pause/resume/cancel paths exist; cancel confirms. Chamber light is runtime verified.
Files / print start	UI scaffold only	Non-destructive SD mount is implemented; media confirmation, indexing, transfer, queueing, and print initiation remain pending.
Motion / thermal / fan	Intentionally disabled	Visible design scaffolding is rejected by the transport until safely validated.
Camera / AI vision	P1 camera working; AI planned	On-demand encrypted P1 liveview is implemented; no vision model or Jetson integration exists.
AMS / filament	Planned	No AMS parser or control workflow is present.
Multi-printer / remote	Planned	One P1S profile today; optional Nova backend/dashboard is a roadmap item.
Diagnostics	Partial	Heap, stack, reset, uptime, coredump, OTA, auth, and connection state are reported; long-duration and full error/HMS work remains.

4 HARDWARE AND SOFTWARE ARCHITECTURE

Guition JC4880P443: ESP32-P4NRW32, 32 MB PSRAM, 16 MB flash, ESP32-C6 Wi-Fi, 4.3-inch ST7701S MIPI-DSI display, GT911 touch, and SDMMC. Firmware: ESP-IDF 6.0.2 / FreeRTOS / LVGL 9.5. Working printer transport: authenticated local TLS/MQTT with CA and printer-identity checks.



Solid lines are implemented; dashed lines are planned. No credentials or private identifiers are shown.



Enclosure or hardware development.

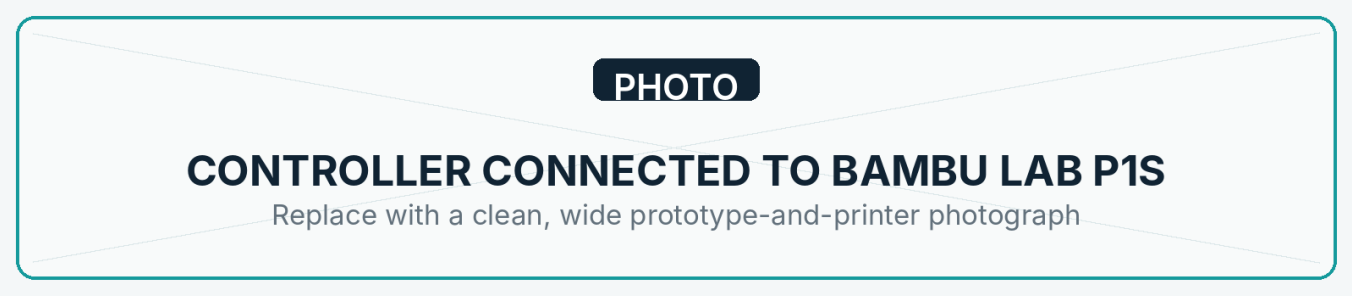
5 WHY ADDITIONAL PRINTERS ARE NEEDED

One P1S can validate only one printer family and one set of firmware, telemetry, camera, accessory, and command behaviors. It cannot establish whether the same fields, responses, reconnect behavior, error states, or interface assumptions hold across Bambu Lab's X and H families.

RECOMMENDED LOAN UNIT	TESTING VALUE
X1 Carbon + AMS	Adds X-series authentication/firmware coverage and a higher-resolution chamber camera plus Micro Lidar behavior—useful for separating model capabilities from P1S assumptions.
H2D + AMS 2 Pro	Adds the broadest H-series protocol challenge: dual-nozzle state, newer camera/sensor roles, active chamber functions, and newer filament-system behavior.

With the existing P1S, these units would provide broad P/X/H family coverage—not certification of every Bambu Lab model or SKU. A-series and per-model certification would remain separate work.

Official references: P1S specification / X1 Carbon technical specification / H2D technical data sheet



Controller connected to Bambu Lab P1S.

6 PROPOSED PRINTER-LOAN ARRANGEMENT

I am requesting one or two printers on a temporary development loan for compatibility development, validation, demonstrations, and preparation of accurate project media. The equipment would be kept in a clean, controlled development environment and used for normal printing, protocol observation, regression testing, and documented demonstrations. The planned LAN-integration work does not require permanent printer modification; I would follow applicable operating and maintenance guidance and document relevant configuration changes.

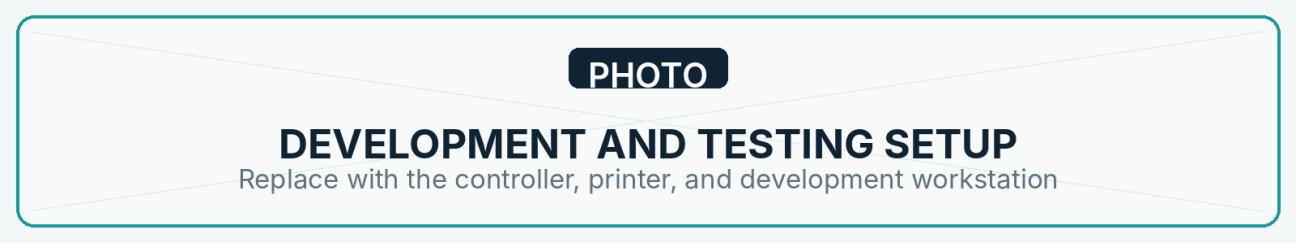
I am willing to comply with reasonable loan terms, provide periodic progress reports, and share compatibility findings and relevant technical feedback. The printers can be returned at the end of an agreed loan period unless a different arrangement is mutually agreed. This request does not presume approval, endorsement, sponsorship, or partnership.

7 TESTING AND VALIDATION PLAN

Complete = observed on the current prototype / Partial = implemented or observed with remaining tests / HW = requested printer required

TEST AREA	P1S	X1C + AMS	H2D + AMS 2 PRO
Discovery / connection	Partial: manual profile; no discovery	HW	HW
Authentication / TLS / MQTT	Complete for current profile	HW	HW
Job / temperature telemetry	Partial: live; regression ongoing	HW	HW + dual-nozzle
Commands / safety gates	Light complete; job controls pending	HW	HW
File handling / print launch	Pending implementation	Pending	Pending
Camera behavior	Partial: local P1 liveview works	HW	HW + camera roles
AMS / filament state	Pending implementation	HW	HW / AMS 2 Pro
Errors / reconnection	Partial code paths; fault tests pending	HW	HW
Firmware / long-duration	Pending matrix	HW	HW
UI adaptation / regression	P1S-focused UI working	HW	HW

Safety-sensitive commands will be exercised only with an appropriate setup, confirmation behavior, and a recoverable test job.



Development and testing setup.

8 DEMONSTRATION AND PROMOTIONAL PLAN

After a reliable demonstration stage, I plan to prepare a dedicated project website, development updates, prototype demonstrations, product walkthroughs, compatibility demonstrations, and educational technical content. Outreach would include multiple 3D-printing and technology YouTube creators, social media content, and carefully targeted Meta advertising.

These are planned activities—not existing creator partnerships or guaranteed coverage. Bambu Lab would be acknowledged as a project supporter only if Bambu Lab agrees and approves the form of that acknowledgment.

9 POTENTIAL VALUE TO BAMBU LAB

A successful independent controller could create additional community-driven use cases while generating practical compatibility feedback across printer families. Potential—not guaranteed—value includes:

- More technical and demonstration content featuring the Bambu Lab ecosystem.
- Engagement from advanced users, makers, and integration-oriented customers.
- Earlier identification and documentation of cross-model compatibility issues.
- Demonstrations of responsible local integration and command gating.

10 DEVELOPMENT ROADMAP

PHASE	SCOPE / EXIT CONDITION
Current — P1S prototype	Hardware-regression test deployed v0.7.1 telemetry, reconnect, camera lifecycle, SD refresh, health data, and guarded actions.
Next technical milestone	Validate supported job commands; harden credentials; implement real local file browse/print start.
Multi-printer compatibility	Add model capability profiles; characterize X1 Carbon and H2D telemetry, cameras, AMS, commands, and UI.
Reliability / hardening	Reconnect, malformed/error-state, firmware-regression, long-duration, and safe-command testing.
Creator demonstration	Presentable enclosure, repeatable demos, accurate photography/video, compatibility summary.
Public launch preparation	Website, walkthroughs, support documentation, disclosures, and outreach materials.

No completion dates are asserted; progress depends on hardware access and compatibility findings.

11 CONTACT AND NEXT STEP

I would welcome a discussion about whether Bambu Lab would consider a temporary development loan of an X1 Carbon and/or H2D, ideally with their corresponding AMS systems.

DIMA	Email: [INSERT EMAIL] / Phone: [INSERT PHONE]
PROJECT	Website: [INSERT PROJECT WEBSITE]
LINKS	Repository: [INSERT REPOSITORY, IF SHARED] / Social: [INSERT SOCIAL LINKS]

Nova Horizon Controller is an independent development project. Bambu Lab is not currently affiliated with or responsible for it.